

React vs Angular — Technical Comparison

1. Introduction

React and Angular are two of the most widely used tools for building the frontend of web applications. React is a JavaScript library from Meta that focuses on building user interfaces out of reusable components, while Angular is a full framework from Google that ships with most of what an application needs out of the box. This document compares them on the points that matter when choosing one for a project.

2. React vs Angular — Comparison

Aspect	React	Angular
Type	JavaScript library for UI	Full frontend framework
Architecture	Component-based; view layer only, other pieces added as needed	Component-based MVC-style structure with modules, services and dependency injection
Language	JavaScript or TypeScript (JSX)	TypeScript (required)
Component Structure	Functions with hooks; markup written in JSX	Classes with decorators; separate template, style and TS files
Data Binding	One-way; updates go through state setters	Two-way available (ngModel); signals and one-way also supported
State Management	Built-in hooks and Context; Redux, Zustand etc. for larger apps	Services with RxJS; signals; NgRx for larger apps
Routing	Not included; React Router or similar	Built-in Angular Router
Forms	Manual or via libraries (React Hook Form, Formik)	Built-in template-driven and reactive forms
Learning Curve	Easier to start; gets harder with the surrounding tools	Steeper at first (TypeScript, RxJS, DI), but consistent afterwards
Performance	Virtual DOM; fast, depends on how components are written	Change detection with zone.js or signals; comparable in most apps
Ecosystem	Very large; many third-party options to pick from	Smaller but official; most tooling comes from the Angular team
Flexibility	High; team decides the structure and libraries	Lower; follows Angular's opinionated conventions
Typical Use Cases	Startups, dynamic UIs, SPAs, Next.js sites	Large enterprise apps, dashboards, long-lived team projects

3. Key Differences

- **Scope:** React only handles the view layer, so routing, forms and state handling are chosen separately. Angular includes all of these by default.
- **Structure:** Angular enforces a fixed project structure, which keeps large teams consistent. React leaves this to the team, which is flexible but can lead to different styles across projects.
- **Language:** Angular requires TypeScript. React works with plain JavaScript and TypeScript is optional, though commonly used.
- **Templates:** React mixes markup and logic in JSX, while Angular keeps HTML templates separate from component code.
- **Learning effort:** React is quicker to pick up initially. Angular takes longer because of concepts like dependency injection and RxJS.
- **Decision-making:** With React, most of the effort goes into choosing libraries. With Angular, most of those decisions are already made by the framework.

4. Conclusion

Neither tool is better in every situation. React tends to suit smaller teams, fast-moving projects, and cases where a flexible setup or a framework like Next.js is wanted. Angular is a reasonable choice for larger applications with many developers, where a fixed structure and built-in tooling help keep the codebase consistent over time. The team's existing experience and the size of the project are usually the deciding factors.